

PUUMALA KK URHEILUKENTTA, Finland

WMO: 027180

Lat: 61.5225N

Lon: 28.1850E

Elev: 322

StdP: 14.53

Time Zone: 2.00 (EUE)

Period: 09-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-10.6	-5.4	-16.4	2.3	-10.5	-10.4	3.2	-5.3	18.9	31.5	18.2	29.9	3.3	250	N/A

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	14.5	82.3	66.0	78.8	64.9	75.4	62.8	68.4	78.6	66.6	75.6	64.8	73.3	7.2	200

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
64.8	93.0	72.9	63.0	87.2	71.4	61.1	81.4	69.6	32.8	79.0	31.4	75.1	30.0	73.4	73.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 17.9	16.1	14.4	DB	-12.5	84.5	8.1	5.3	-18.3	88.3	-23.0	91.4	-27.6	94.3	-33.5	98.2	(4)
(5)			WB	-12.7	70.4	7.5	2.7	-18.1	72.3	-22.5	73.8	-26.8	75.4	-32.2	77.3	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	41.1	16.6	21.8	27.6	38.4	51.5	58.8	65.1	61.6	52.9	40.1	32.9	24.7	(6)
(7)		DBStd	17.97	12.47	11.02	8.15	6.18	7.99	6.72	7.12	5.70	5.85	6.35	7.29	10.88	(7)
(8)		HDD50	4604	1035	790	693	350	79	9	0	1	40	310	513	784	(8)
(9)		HDD65	8871	1500	1210	1158	798	422	207	91	137	364	772	963	1249	(9)
(10)		CDD50	1356	0	0	0	2	125	273	467	360	125	4	0	0	(10)
(11)		CDD65	150	0	0	0	0	4	21	94	31	0	0	0	0	(11)
(12)		CDH74	1043	0	0	0	0	53	157	662	171	0	0	0	0	(12)
(13)		CDH80	220	0	0	0	0	2	23	159	36	0	0	0	0	(13)
(14)	Wind	WSAvg	7.5	7.3	8.1	8.2	7.1	6.8	7.7	6.2	6.8	7.5	8.1	8.1	8.1	(14)
(15)	Precipitation	PrecAvg	25.3	2.0	1.5	1.4	1.2	1.6	2.4	3.1	2.8	2.0	2.7	2.4	2.2	(15)
(16)		PrecMax	29.4	3.3	3.5	2.9	2.6	3.6	3.9	5.9	5.4	4.9	6.1	5.1	3.7	(16)
(17)		PrecMin	19.1	0.4	0.2	0.3	0.4	0.4	0.9	1.0	0.9	0.9	0.6	0.4	0.7	(17)
(18)		PrecStd	2.7	0.7	0.8	0.6	0.6	0.9	0.9	1.3	1.4	1.1	1.3	1.1	0.9	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	37.3	41.1	48.1	64.4	79.3	83.0	88.7	84.4	68.7	55.8	49.8	45.4	(19)
MCWB			35.8	36.8	38.5	49.0	59.7	63.7	70.0	68.3	60.0	50.9	46.7	44.1	(20)	
2%		DB	35.6	37.7	43.7	56.9	74.5	78.4	84.3	78.5	65.7	53.1	46.1	40.5	(21)	
		MCWB	34.7	35.2	37.1	45.6	57.7	62.2	67.2	65.5	58.7	50.1	44.5	38.9	(22)	
5%		DB	33.3	35.5	40.0	52.2	70.1	74.5	80.7	74.4	63.2	51.3	43.9	38.0	(23)	
		MCWB	32.4	34.1	34.9	42.9	55.1	60.7	66.0	62.5	57.0	48.7	42.6	36.5	(24)	
10%		DB	32.1	33.8	37.2	48.8	65.9	70.9	77.5	71.0	60.9	49.4	41.6	35.8	(25)	
		MCWB	31.3	32.9	33.6	40.9	53.3	59.5	65.5	61.5	56.2	47.0	40.4	34.6	(26)	
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	36.3	37.4	40.3	49.7	63.0	67.6	71.8	69.8	61.7	52.7	47.2	44.2	(27)
MCD B			36.9	39.4	45.1	58.3	74.3	79.1	83.8	81.9	66.2	54.7	48.8	45.2	(28)	
2%		WB	34.8	35.7	37.7	46.7	59.6	64.6	69.4	66.6	59.9	51.2	44.8	39.1	(29)	
		MCD B	35.5	37.3	42.2	56.0	69.5	74.3	80.1	74.4	63.8	52.7	46.2	40.1	(30)	
5%		WB	33.1	34.3	35.9	44.2	57.2	62.1	67.8	64.8	58.5	49.4	42.7	36.8	(31)	
		MCD B	33.7	35.4	39.0	50.6	67.3	71.5	77.6	72.4	61.8	51.0	43.7	37.9	(32)	
10%		WB	31.5	33.1	34.4	42.0	54.8	60.1	66.4	62.6	56.9	47.1	40.6	34.8	(33)	
		MCD B	32.2	34.0	36.6	47.5	64.6	69.2	75.4	69.3	60.5	49.1	41.6	35.7	(34)	
(35)	Mean Daily Temperature Range	5% DB	MDBR	8.9	8.6	11.8	14.6	17.2	15.1	14.5	12.6	9.9	7.2	5.5	6.5	(35)
MCDBR			6.7	7.9	14.4	20.9	24.3	20.8	19.4	17.7	12.8	9.5	7.6	7.0	(36)	
MCWBR			6.5	6.7	9.6	12.0	10.9	9.2	7.5	7.1	7.4	7.3	7.4	7.1	(37)	
5% WB		MCDBR	6.1	7.5	11.8	18.1	20.8	18.4	17.9	16.3	11.3	8.7	7.4	7.1	(38)	
		MCWBR	6.0	6.6	8.6	11.1	10.6	9.5	7.8	7.4	7.2	7.6	7.3	7.3	(39)	
(40)	Clear-Sky Solar Irradiance	taub	0.248	0.293	0.320	0.360	0.356	0.353	0.377	0.362	0.338	0.320	0.270	N/A	(40)	
(41)		taud	2.057	2.127	2.204	2.344	2.440	2.480	2.428	2.454	2.511	2.407	2.147	N/A	(41)	
(42)		Ebn at Noon	141	210	249	263	275	278	266	261	248	206	132	N/A	(42)	
(43)		Edh at Noon	12	23	30	32	31	31	32	29	23	18	11	N/A	(43)	
(44)	All-Sky Solar Radiation	RadAvg	73	252	715	1161	1582	1672	1628	1254	723	293	79	37	(44)	
(45)		RadStd	12	61	100	125	185	106	159	152	83	44	13	4	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (35 neighbors)	+0.92	N/A	N/A	N/A	N/A	N/A	-310	N/A	N/A	N/A

Nomenclature: See separate page